

Math 242, S15
Quiz 4

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer..

Problem 1. (5 points) Find a general solution using the substitution method for the differential equation:

$$xy^2y' = x^3 + y^3.$$

(Hint: using the substitution $v = y/x$)

Solution: $y' = \frac{x^2}{y^2} + \frac{y}{x}$

let $v = \frac{y}{x}$, i.e. $y = vx$ then

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus we have.

$$x + x \frac{dv}{dx} = \frac{1}{v^2} + x$$

$$x \frac{dv}{dx} = \frac{1}{v^2}$$

$$v^2 dv = \frac{1}{x} dx \Rightarrow \int v^2 dv = \int \frac{1}{x} dx$$

$$\Rightarrow \frac{v^3}{3} = \ln|x| + C_1$$

$$\frac{y^3}{x^3} = 3\ln|x| + C$$

$$y^3 = x^3(3\ln|x| + C)$$

$$y = x(3\ln|x| + C)^{\frac{1}{3}}$$

Problem 2. (5 points) Verify whether the given differential equation is exact and then solve it if yes:

$$(4x - y) dx + (6y - x) dy = 0.$$

Solution: $M(x, y) = 4x - y$, $N(x, y) = 6y - x$

$$\left. \begin{array}{l} \frac{\partial M}{\partial y} = -1 \\ \frac{\partial N}{\partial x} = -1 \end{array} \right\} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{exact!}$$

$$\frac{\partial F}{\partial x} = M, \quad \frac{\partial F}{\partial y} = N$$

$$\begin{aligned} \Rightarrow F(x, y) &= \int M(x, y) dx \\ &= \int (4x - y) dx + g(y) \\ &= 2x^2 - xy + g(y) \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial y} = N &\Rightarrow \frac{\partial}{\partial y}(2x^2 - xy + g(y)) = 6y - x \\ -x + g'(y) &= 6y - x \end{aligned}$$

$$g'(y) = 6y$$

$$\Rightarrow g(y) = \int 6y dy = 3y^2$$

$$\text{Thus } F(x, y) = 2x^2 - xy + 3y^2$$

and the general solution of the equation is

$$2x^2 - xy + 3y^2 = C$$